



Photonfocus AG

Photonfocus SDK for Linux

Quickstart guide

1 PFSDK for Linux quickstart guide

After extracting the tarball (in which this guide is included) the following directory structure can be found.

- *bin*: contains any executables packaged with the PFSDK.
- *doc*: documentation for the different APIs.
- *driver*: network filter driver for the Linux Kernel.
- *examples*: sample projects showing SDK usage.
- *include*: library header files.
- *lib*: library object files.

The first thing the user should do is decide whether to install the filter driver. Installing the filter driver will provide better performance. Whether the filter driver is used or not can be configured in the API.

1.1 Installing the filter driver

Kernel modules need to be recompiled for each minor version change in the kernel. Hence, providing a driver binary for all kernel versions would not be feasible. For convenience, a DKMS install is provided, which will install the driver source and recompile it every time there is a kernel change.

1.1.1 DKMS install

This requires the system to have DKMS installed. In an Ubuntu distribution, DKMS may be installed with the following command:

```
sudo apt-get install dkms
```

After installing DKSM, execute the script found in the *dkms* folder with root privileges.

```
sudo ./install_with_dkms.sh
```

1.2 Setting up the network adapter

Find the network adapter that the camera is connected, do this by using *ifconfig* or *ip address show*

1.2.1 Enable Jumbo frames (MTU)

`sudo ip link set ens160 mtu 9014` . Replace *ens160* with the name of the network interface where the camera is connected

1.2.2 Enable link-local (if applicable)

If the camera is connected to a larger dhcp network skip this step

In the Ubuntu default desktop go to Settings -> Network select the network adapter and set the link-local option in the Ipv4 and Ipv6 tabs. Doing it via console it's also accepted.

Ethernet (ens160)
+

Connected - 10000 Mb/s



Ethernet (ens192)
+

Connected - 10000 Mb/s



Ethernet (ens224)
+

Connected - 10000 Mb/s



Intel Ethernet
+

Connected - 1000 Mb/s



Cancel
Wired
Apply

Details
Identity
IPv4
IPv6
Security

IPv4 Method
☐ Automatic (DHCP)
☒ Link-Local Only
☐ Manual
☐ Disable
☐ Shared to other computers

DNS
Automatic Separate IP addresses with commas

Routes
Automatic 

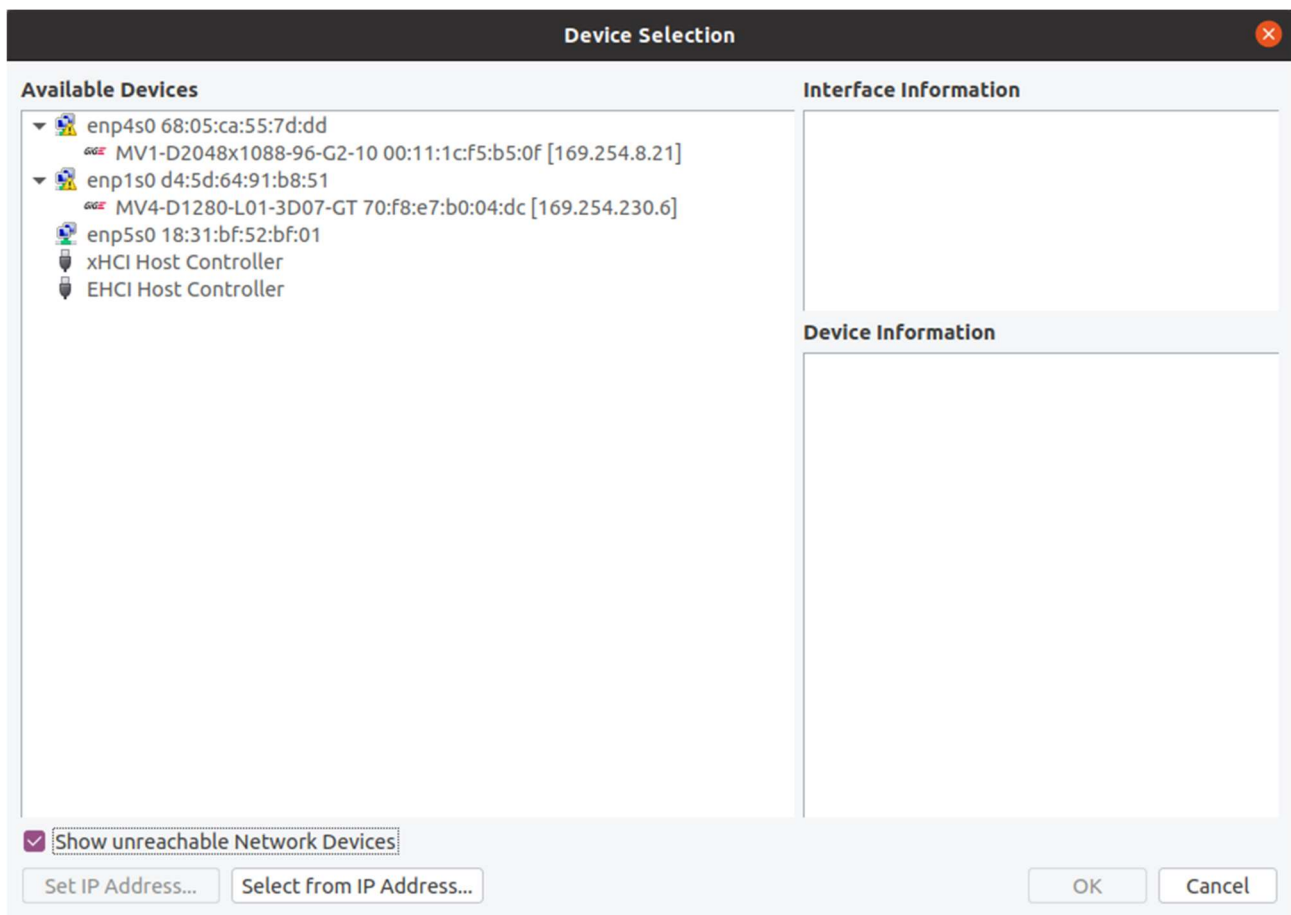
Address	Netmask	Gateway	Metric	
				

1.2.3 NIC identification issue

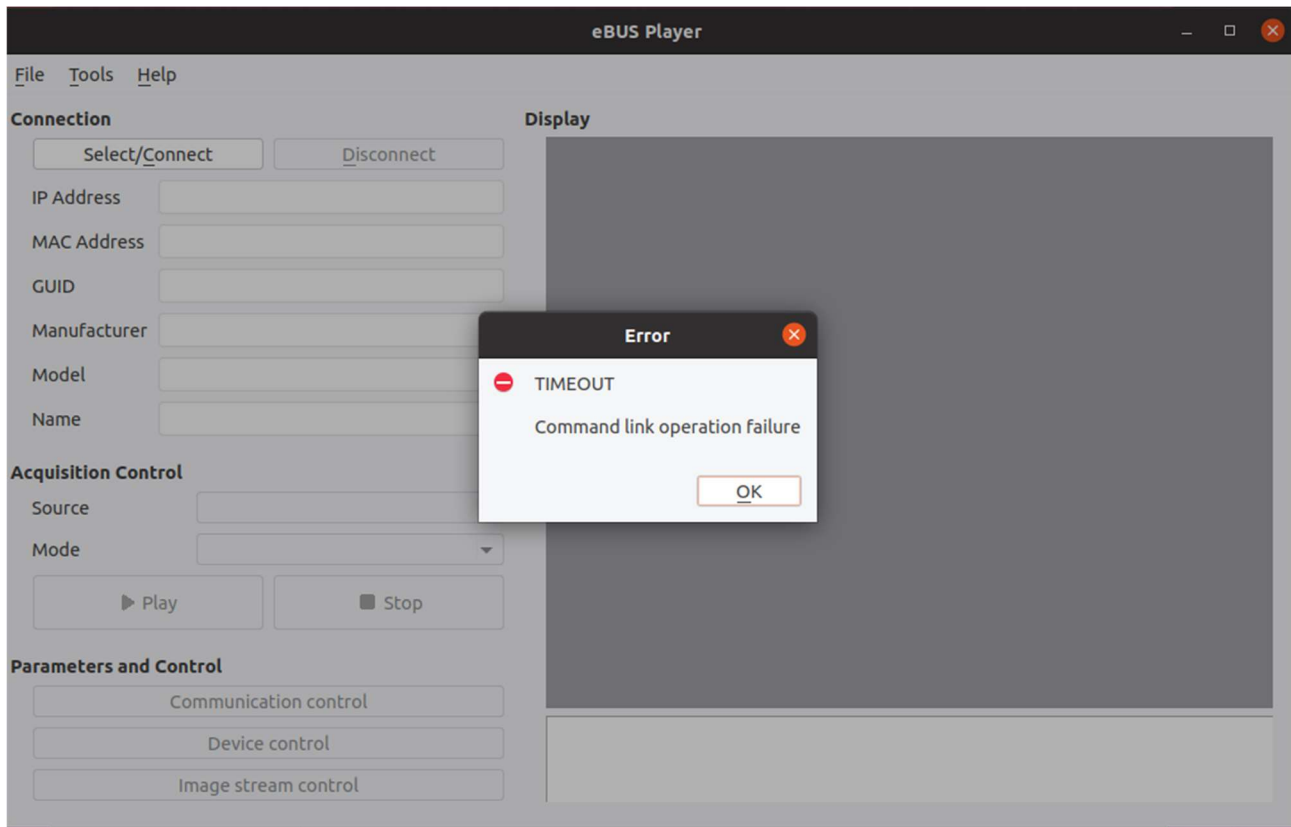
This issue may happen when the user has two cameras connected to different Network Cards. It can happen that Linux assigns the same IP Address (first 2 bytes of the entire address) for the two NICs reachable. As a result of this card identification is wrong and therefore connection with one of the cameras always fails.

```
xavier@Xavier-PC:~$ ip address show dev enp4s0
3: enp4s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9000 qdisc fq_codel state UP group default qlen 1000
    link/ether 68:05:ca:55:7d:dd brd ff:ff:ff:ff:ff:ff
    inet 169.254.105.158/16 brd 169.254.255.255 scope link noprefixroute enp4s0
        valid_lft forever preferred_lft forever
    inet6 fe80::4441:9de8:b866:e5e9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
xavier@Xavier-PC:~$ ip address show dev enp1s0
4: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether d4:5d:64:91:b8:51 brd ff:ff:ff:ff:ff:ff
    inet 169.254.213.35/16 brd 169.254.255.255 scope link noprefixroute enp1s0
        valid_lft forever preferred_lft forever
    inet6 fe80::356e:e9b4:eb04:e0ee/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

enp4s0 and enp1s0 share same partial address (first two bytes) 169.254.XXX.XXX.

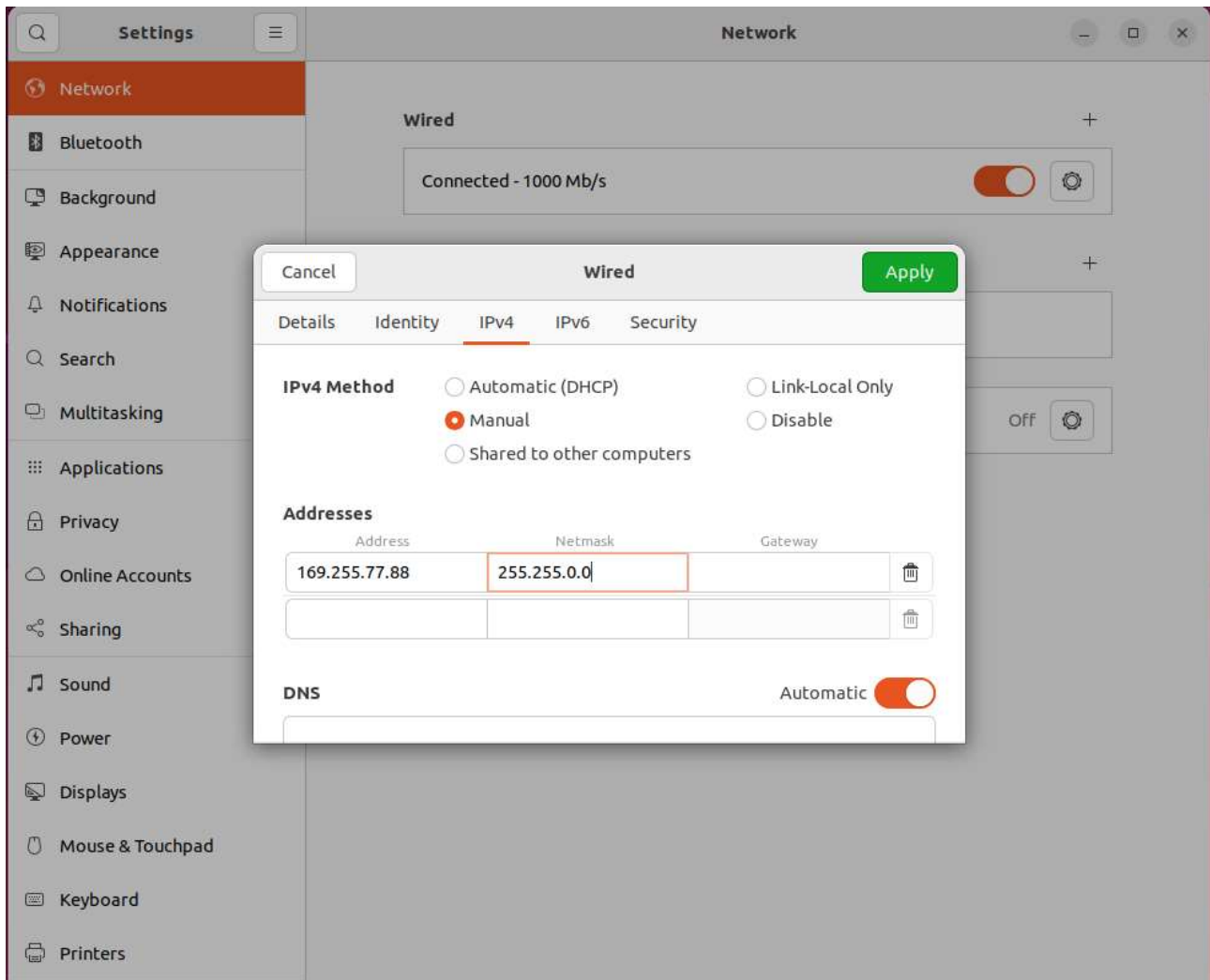


Issues with Network Identification in eBUS Player



Failed Connection with MV1 Camera

There is one possible workaround to this kind of situation, the trick is to assign manually a specific IP address to one of the NICs to differentiate both cards between them. You can do so in Ubuntu by setting Network properties and configuring IPv4 to Manual method and setting the new address that will be assigned.



Setting Specific NIC IP Address

Now each Network Card has different IP Address and Discovery process is right (see *Discovery Status*). Connection for each camera can be done as usual.

```
xavier@Xavier-PC: ~/Development/PFSDK_2023.2.0_Linux64-Packaged/examples/C++/DiscoverCo...
xavier@Xavier-PC:~/Development/PFSDK_2023.2.0_Linux64-Packaged/examples/C++/DiscoverConfigAndGrab_Console/bui
ld$ ./PFCameraLib_DiscoverConfigAndGrab_Console

Cameras found:

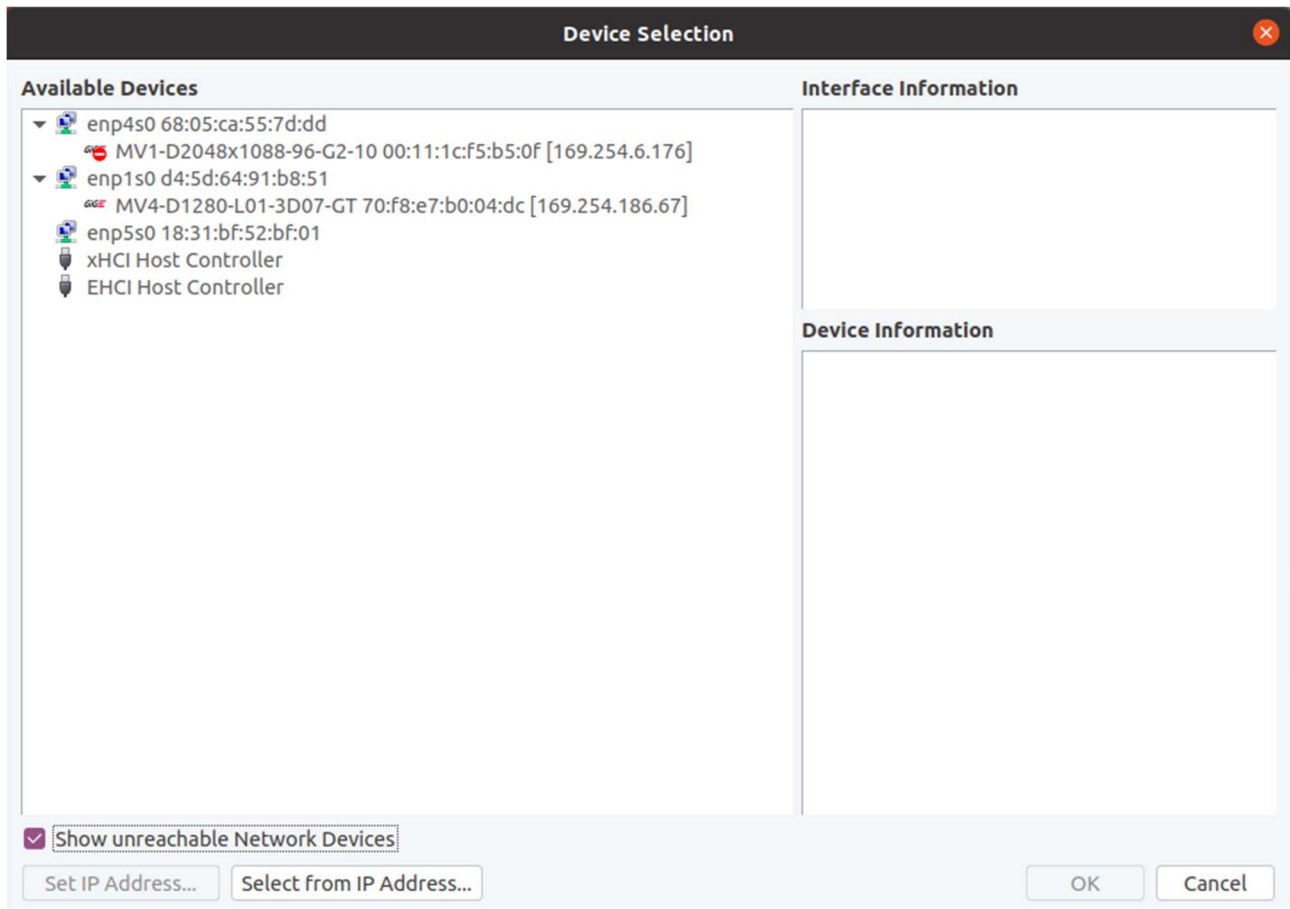
1- MV1-D2048x1088-96-G2-10
  Model Name = MV1-D2048x1088-96-G2-10
  Manufacturer Info = Photonfocus AG
  Version = Version 1.0 (02.05.03)
  Serial Number = 041000027214
  User Defined Name = MV1 SOLPI
  MAC = 00:11:1C:F5:B5:0F
  IP = 169.255.70.80
  Subnet Mask = 255.255.0.0
  Default Gateway = 0.0.0.0
  Discovery Status = OK

2- MV4-D1280-L01-3D07-GT
  Model Name = MV4-D1280-L01-3D07-GT
  Manufacturer Info = Photonfocus AG
  Version = Version 1.0
  Serial Number = 053700049120
  User Defined Name = Device01
  MAC = 70:F8:E7:B0:04:DC
  IP = 169.254.252.113
  Subnet Mask = 255.255.0.0
  Default Gateway = 0.0.0.0
  Discovery Status = OK

Select a camera from the list: █
```

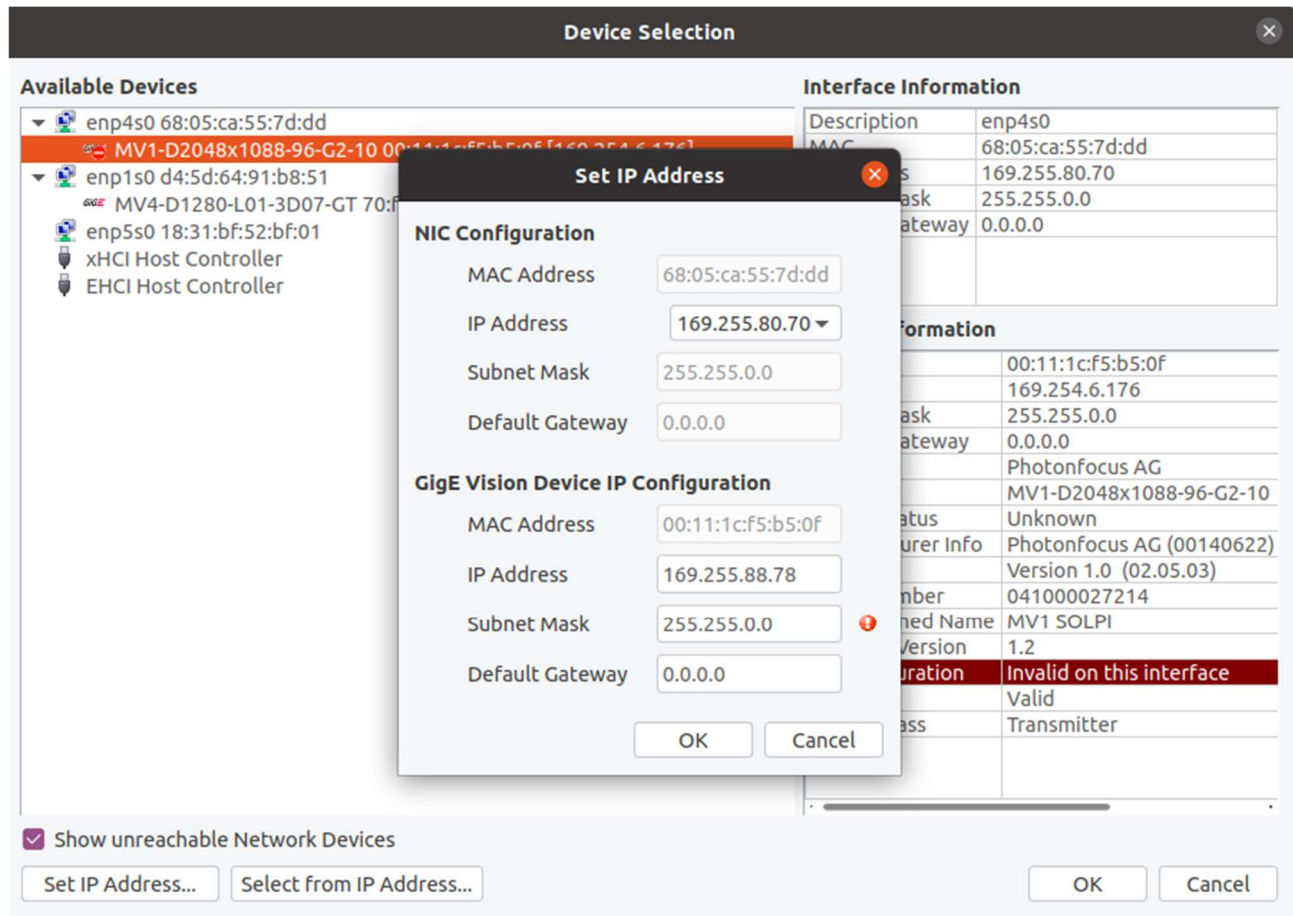
PFSDK Discovery Status

Note that eBUS Player or other GEV applications could require to reassign the IP Address for the connected Camera which his NIC Address has been modified.



Wrong IP Address for camera MV1

In eBUS Player you can do so by setting IP Address and assigning the NIC Configuration appropriate. After that you can change the IP Address for the device.



Set IP Address for the MV1 Camera

With this setting both connections are ready and can be used as usual.

Device Selection

Available Devices

enp4s0 68:05:ca:55:7d:dd

MV1-D2048x1088-96-G2-10 00:11:1c:f5:b5:0f [169.255.88.78]

enp5s0 18:31:bf:52:bf:01

enp1s0 d4:5d:64:91:b8:51

MV4-D1280-L01-3D07-GT 70:f8:e7:b0:04:dc [169.254.186.67]

xHCI Host Controller

EHCI Host Controller

Interface Information

Description	enp4s0
MAC	68:05:ca:55:7d:dd
IP Address	169.255.80.70
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0

Device Information

MAC	00:11:1c:f5:b5:0f
IP	169.255.88.78
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
Vendor	Photonfocus AG
Model	MV1-D2048x1088-96-G2-10
Access Status	Open
Manufacturer Info	Photonfocus AG (00140622)
Version	Version 1.0 (02.05.03)
Serial Number	041000027214
User Defined Name	MV1 SOLPI
Protocol Version	1.2
IP Configuration	Valid
License	Valid
Device Class	Transmitter

☐ Show unreachable Network Devices

Set IP Address... Select from IP Address...

OK Cancel

Ready to Connect

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1.3 Building and running the samples

The samples are provided with a CMake configuration file. Install CMake 3.15 or greater before proceeding. The following pattern can be used to configure, build, and run the samples:

```
cd examples/C++/DiscoverConfigAndGrab_Console
mkdir build
cd build
cmake ..
make
./PFCameraLib_DiscoverConfigAndGrab_Console
```

If the *examples* folder is moved in relation to the *lib* folder and the SDK was not extracted into one of the standard CMake search paths, additional environment variables may need to be specified.

Visit the following link for more information about CMake's `find_package`:

https://cmake.org/cmake/help/latest/command/find_package.html#search-procedure

1.4 Read the documentation

The documentation is provided in HTML format. Use a browser to navigate the documentation:

- `doc/PFCameraLib/html/index.html` (C++ API)
- `doc/PFPyCameraLib/index.html` (Python API)